A Critical Study on Sustainable Nature of Solar Panels

Avula Ajay, Department of Electrical Engineering, IIT Kanpur
Mentored by : Dr. Suvendu Samanta, Department of EE, IIT Kanpur

Introduction

- With rising global energy demands and growing concerns about climate change caused by the burning of fossil fuels, solar panels have emerged as a leading source of green energy.
- However, despite their renewable origin, emission-free operation, and low cost, the overall sustainability of solar panels depends on their entire life cycle-from raw material extraction to final disposal.
- While solar panels appear to be a remarkable source of green energy, they come with hidden limitations that are frequently overlooked and cause significant environmental impact.



Research Question

- Are solar panels truly green source of energy?
- How solar panels cause damage to the nature?
- What are the hidden environmental concerns raised by large-scale installation of solar panels?
- How can sustainability of solar panels be improved?

Methodology

- This study adopts a qualitative research methodology based on a comprehensive review of published research articles, life cycle assessment (LCA) reports, and global energy databases.
- A cradle-to-grave life cycle assessment (LCA) framework is used to evaluate the sustainability of solar PV systems across:
  1. Manufacturing
  2. Transportation & installation
  3. Operation (electricity generation)
  4. End-of-life recycling & disposal
- The study compares solar PV to fossil fuels on key metrics:
  1. Carbon emissions
  2. Energy payback time (EPBT)
  3. Lifetime energy output
- It also highlights frequently less-discussed issues:
  1. Emissions during manufacturing
  2. Toxic and rare material waste
  3. End -of-life disposal
  4. Solar Heat Island Effect (SHIE)
  5. Biodiversity and climate impacts

No primary experiments were conducted; findings are based completely on secondary data.

Literature Review

Sustainability of Solar Panels-Benefits

- Solar PV offers emission-free electricity production and reduces fossil fuel dependence.
- Fthenakis & Kim (2009): Life cycle emissions of mono-Si panels , coal and Natural gas are 40–60g CO_2/kWh , coal 800–1000g CO_2/kWh and 500 CO_2/kWh respectively.
- IEA PVPS (2020): Technological advancement have reduced EPBT to 1–2 years, with panel lifespans >25 years.

Underreported Sustainability Concerns

1. Manufacturing Emissions
IRENA (2022), Held et al. (2021):
Silicon purification and module assembly are energy-intensive, often powered by coal (especially in China).
➤ Can contribute >75% of total life cycle emissions.

2. Toxic Materials & Waste
- Cucchiella et al. (2015), Kang et al. (2014):

Panels contain cadmium, lead, antimony → risk to soil and water.
- Lu et al. (2015), Fthenakis (2009):

Silicon purification releases $SiCl_4$, a toxic byproduct. Poor disposal has led to local environmental damage.

3. End-of-Life Waste
IRENA (2016):
Projected 78 million tonnes of panel waste by 2050.
➤ Recycling is costly and inefficient (only 60–70% recovery).
➤ Highlights the need for closed-loop systems and policy reform (EPR).

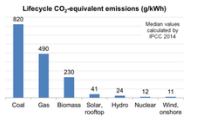
4. Solar Heat Island Effect
Barron-Gafford et al. (2016):
Large solar farms increase local temperatures (Photovoltaic Heat Island Effect).
➤ Alters temperature, humidity, and plant growth in deserts/semi-arid areas.

5.Ecological Impacts
Hernandez et al. (2014):
Solar farms can fragment habitats, displace wildlife, and degrade ecosystems — even in deserts.
➤ Ecological footprint often ignored in national policies.

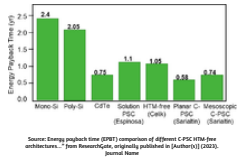
6. Global Climate Alterations
Widespread deployment of solar farms may influence cloud dynamics, atmospheric circulation, and surface energy balance, potentially leading to large-scale climate feedbacks.
➤ Highlights the need to assess global environmental trade-offs of solar expansion.

Discussion

Positive Sustainability Metrics



Source: Solar.com (2024)



Source: Energy payback time (EPBT) comparison of different C-PSC HTM-free architectures..." from ResearchGate, originally published in [Author(s)] (2023), Journal Name

- Lifetime emissions:
  Monocrystalline PV → 40–60g CO_2/kWh
  vs. Coal → 800–1000g CO_2/kWh
- Energy Payback Time:
  Modern panels = 1–2 years
  Lifespan = 25+ years

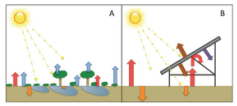
➡ Solar PV is clearly greener in the use phase.

Sustainability Drawbacks

a) Manufacturing Emissions
- Up to 80% of emissions occur before installation
- High-temperature, coal-powered silicon processing

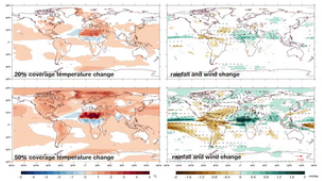
b) Toxic Waste & Rare Materials
- Contains lead, cadmium, antimony, silver
- $SiCl_4$ byproduct → pollutes water/soil if mismanaged
- Mining (e.g., quartz, indium) harms local ecosystems

c) End-of-Life Waste
- By 2050 → 78 million tonnes of panel waste
- Current recycling recovers only low-value materials
- Rest often ends in landfills or incineration



d) Solar Heat Island Effect (SHIE)
- Large farms raise local temperatures by 2–5°C
- Alters microclimates, harms flora & insects

e) Biodiversity & Land Use Causes:
- Habitat fragmentation
- Species displacement
- Migratory disruption
- Water cycle disturbance in arid zones



f) Global Climate Alterations
- Studies (e.g., Millstein & Menon, 2016) suggest:
- Changes to wind, clouds, surface albedo

Conclusion

- Although solar energy is a key pillar of the global green energy transition, it is not completely green. Its large-scale deployment can lead to issues such as land-use conflicts, lifecycle emissions, toxic panel waste, global climate alterations, and the Solar Heat Island Effect.
- Addressing these challenges requires climate-sensitive siting, circular production, and robust recycling systems. Integrating efficient storage solutions and promoting community-inclusive policies are also essential. By adopting these improvements, solar energy can move closer to becoming a truly sustainable and responsible energy source.

References

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